INCH-POUND

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MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, 11747101
(RANGE AND RANGE RATE GENERATOR)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of Range and Range Rate Generator printed circuit board, hereinafter referred to as range and range rate generator, which forms a part of SCG Servo Test Set 11747100 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electric and Electronic Assemblies

(Unless other indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10554938 - Connector
11747101 - Generator, Range and Range Rate
11747103 - Schematic, Generator, Range and Range Rate

PACKAGING DATA SHEET

PS11747101 - Packaging of Generator, Range and Range Rate

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11747101, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the range and range rate generator shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
co Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The range and range rate generator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The range and range rate generator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The range and range rate generator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The range and range rate generator shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

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3.3.4 Temperature.

3.3.4.1 Operating. The range and range rate generator shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between 32 degrees F and +131 degrees F.

3.3.4.2 Non-operating (storage). The range and range rate generator shall not suffer permanent damage when subjected to the constant temperature extremes of -80 degrees F for a period of 24 hours and +185 degrees F for a period of 4 hours after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The range and range rate generator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be electrically continuous as specified on schematic diagram 11747103.

3.4.2 Resistance. Resistance shall be 1000 ohms + 10% between the following terminals.

From	To
P1-11	P1-30
P1-45	P1-30

3.5 Performance.

3.5.1 Input signal. For the requirements of 3.5.3 through 3.5.8, logic input and output levels shall be as follows:

Logic 0: 0.0 ± 0.5 Vdc Level

Logic 1: 4.0 ± 1.2 Vdc Level

: Momentary Transition from Logic 1 to Logic O

3.5.2 Input power. For the requirements of 3.5.3 through 3.5.8, input power shall be provided in accordance with the test setup illustrated in Figure 1.

3.5.3 Output pin 3. The output, with a 1K load between pin 3 and pin 22 shall be +12 Vdc ± 5.0 percent between pin 3(+) and pin 22(-).

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3.5.4 Range output. With a Logic O applied to P1-11 and with P1-45, P1-5, P1-6 at logic O, the range output pulses shall have the periods specified in Table I. All outputs shall be positive 50 percent square waves from OV to 4.0 ± 1.2 Vdc. P1-24 shall be used as return for all signals.

TABLE I. Range outputs

DESCRIPTION	TP	CHARACTERISTIC	TERMINAL
5 M	1	16.6 ± 0.4 ms	P1-52
10 M	2	33.2 ± 0.8 ms	P1-51
20 M	3	66.4 ± 1.6 ms	P1-55
40 M	4	132.8 ± 3.2 ms	P1-56
80 M	5	265.6 ± 6.4 ms	P1-54
160 M	6	531.2 ± 12.8 ms	P1-53
320 M	7	1.06 ± 0.03 sec	P1-49
640 M	8	2.12 ± 0.06 sec	P1-50
1280 M	9	4.24 ± 0.12 sec	P1-47
2560 M	10	8.48 ± 0.24 sec	P1-46

3.5.5 Range Rate output. With a Logic O applied to P1-45 and with P1-11, P1-5, P1-6 at logic O, the range rate output pulses shall have the periods specified in Table II. All outputs shall be positive 50 percent square wave from OV to 4.0 ± 1.2 Vdc. P1-24 shall be used as return for all signals.

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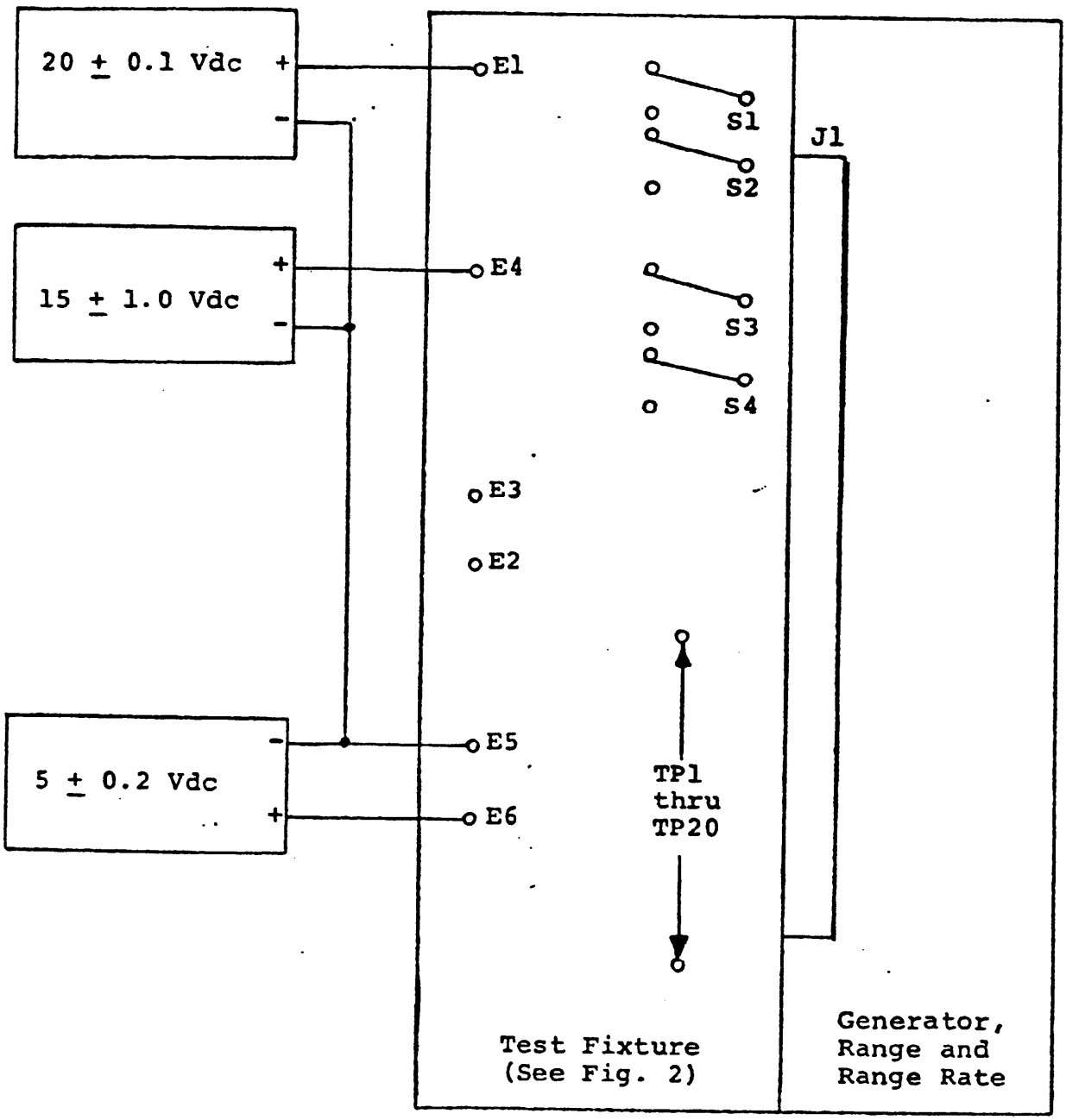


FIGURE 1. Range and range rate generator test.

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TABLE 11. Range rate outputs

DESCRIPTION	TP	CHARACTERISTIC	TERMINAL
2.5 M/S	11	132.8 ± 3.2 ms	P1-19
5 M/S	12	265.6 ± 6.4 ms	P1-17
10 M/S	13	531.2 ± 12.8 ms	P1-15
20 M/S	14	1.06 ± 0.03 sec	P1–41
40 M/S	15	2.12 ± 0.06 sec	P1-43
80 M/S	16	4.24 ± 0.12 sec	P1-40
160 M/S	17	8.48 ± 0.24 sec	P1-39

3.5.6 Range count signals.

3.5.6.1 Up count ranges. With the input Logic Levels applied as specified in Table III, step 1, two consecutive output signals as specified in Table I shall first be out of phase for the first part of the run indicating a count up direction control, and then in phase for the second part of the run indicating a count down direction control. The time for the first part of the run shall be 8.0 ± 0.4 seconds for the 5M and 10M outputs of Table I. The time for an entire run shall be 16.0 ± 0.8 seconds.

TABLE 111. Input logic levels

Step	P1-5	P1-6	P1-11	P1-45
1	0	1	*	1
2	0	1	1	*
3	1	0	*	1
4	1	0	1	*

*Not to be toggled until ready to make run.

3.5.6.2 Down count range. With the input logic levels applied as specified in Table IV, step 3, two consecutive output signals as specified in Table I shall be in phase for the first part of the run, indicating a count down direction control, and then out of phase for the second part of the run indicating a count up direction control. The time for the first part of the run shall be 8.0 ± 0.4 seconds for the 5 M and 10 M outputs of Table I. The time for the entire run shall be 16.0 ± 0.8 seconds.

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3.5.7 **Range rate count signals.**

3.5.7.1 Up count range rate. With the input logic levels applied as specified in Table III, step 2, two consecutive output signals as specified in Table 11 shall-be out of phase for the first part of the run, indicating a count up direction control, and then in phase for the second part of the run, indicating a count down direction control. The time for the first part of the run shall be 8.0 ± 0.4 seconds for the 2.5 M/S and 5 M/S outputs of Table II. The time for the entire run shall be 16.0 ± 0.8 seconds.

3.5.7.2 **Down count range** *rate.* With the input logic levels applied as specified in Table 111, step 4, two consecutive output signals as specified in Table II shall be in phase for the first part of the run, indicating a count down direction control, and then out of phase for the second part of the run, indicating a up count direction control. The time for the first part of the run shall be 8.0 ± 0.4 seconds for the 2.5 M/S and 5 M/S outputs of Table II. The time for the entire run shall be 16.0 ± 0.8 seconds.

3.5.8 **Sign-Sign output.** In 3.5.7.1 and 3.5.7.2, whenever two consecutive output signals of Table II are out of phase, indicating a count up condition, P1-13 shall be a logic 1 and PI-9 shall be a logic o. Whenever two consecutive signals are in phase, indicating a count down condition, PI-9 shall be a logic 1 and P1-13 shall be a logic O.

3.6 **First article.** When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task)F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other or facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. all items shall meet all requirements of sections 3 and 5. The inspections set forth In this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The requirement for first article and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract and the applicable drawing. 'The sample for-the first article tests shall consist of three range and range rate generators manufactured in the same manner, using the same materials, equipment, processes amd procedure as in regular production. All parts and materials, including packing and packaging, shall be obtained from the same source of supply as used in regular production.

4.3.1.1 Government testing. When the Government is responsible for conducting first article tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and the applicable drawings. The contractor shall submit a record of the inspection with the sample including Certified Test Report (CTR) or Certificate of Conformance(COC) for the materials.

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4.3.1.2 Contractor testing. When the Contractor is responsible for conducting first article tests, the sample shall be inspected by the contractor for all the requirements of-the contract, all of the examinations and tests specified in Table IV (Functional Test) and Table V (Special Sampling) herein and to all requirements of the applicable drawings. The sample and a record of this inspection, including CTR/COC, shall be submitted for the Government approval. The Government reserves the right to witness the Contractor's inspection.

4.3.2 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table IV shall be performed on a 100% basis.

TABLE IV. Functional tests

(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
101.	Electrical	3.4	4.7.2
102.	Performance	3.5	4.7.3
Minor:		None	N/A

4.5 Special sampling. One range and range rate generator out of the first 50 produced and one out of each 100 produced thereafter shall-be selected at random. The sample shall have met the requirements and tests of Table IV and shall then meet the tests in Table v. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE V. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity	3.3*5	4.7.1.5
Minor:		None	N/A

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component Inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment Is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table VI shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE VI. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Power Supply	20.0 ± 0.1 Vdc, regulated. AC ripple: Less than 50 mv (millivolt) peak-to-peak. Capacity: 50 mA (milliamp).
2.	Power Supply	15.0 ± 1 Vdc, regulated. AC ripple: Less than 50 mv peak-to-peak. Capacity: 50 mA.
3*	Power Supply	5.0 ± 0.2 Vdc, regulated. AC ripple: Less than so mv peak-to-peak. Capacity: 600 mA.
4.	Oscilloscope	Dual channel. Rise time of scope plus measuring probe: 10 ns (nanoseconds) or less. Sweep period: 50 ns/div. or greater. Input cap: Less than 25 pF. Input res: Greater than 8 megohms.

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TABLE VI. Test equipment list (continued)

ITEM NUMBER		DESCRIPTION
5.	Test Fixture	As specified in Figure 2. See Table VII for test fixture parts list.
6.	Ohmmeter	To measure 1000 ohms ± 10 percent.
7.	Digital Voltmeter (DVM)	A range of at least 0.01 to 100 Vdc. Accuracy of at least ± 0.2 percent of reading. Input impedance of at least 10 megohms on all ranges.
8.	Frequency Counter	Dual input. Time interval B-C function: 1 ms (milliseconds) to 10 sec. Frequency Range: 0.01 Hz to 100 Hz. Input Sensitivity: Minimum 100 mV rms each channel, Variable Attenuator 40 dB.

TABLE VII. Test fixture parts list

DESIGNATION	NOMENCLATURE AND DESCRIPTION
El, E3, E4, E6	Binding post, H.H. Smith, 257-R or equiv.
E2, E5	Binding post, H.H. Smith, 257-B or equiv.
J1	Connector, Part No. 10554938
Sl, S2, S3,	Switch SPDT
TP1-TP20	Tip Jack, H.H. Smith, 1501 or equiv.
R1	Resistor, 4.OK, 0.1%, 1/2 Watt
R2	Resistor, 5.37K, 0.1%, 1/2 Watt
R3	Resistor, lK, 1%, 1/2 Watt

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4.7 Methods of inspection.

4.7*1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The range and range rate generator shall be attached to the vibration generator directly or with a fixture and securely held by its normal means of attachment. After vibration, the range and range rate generator shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

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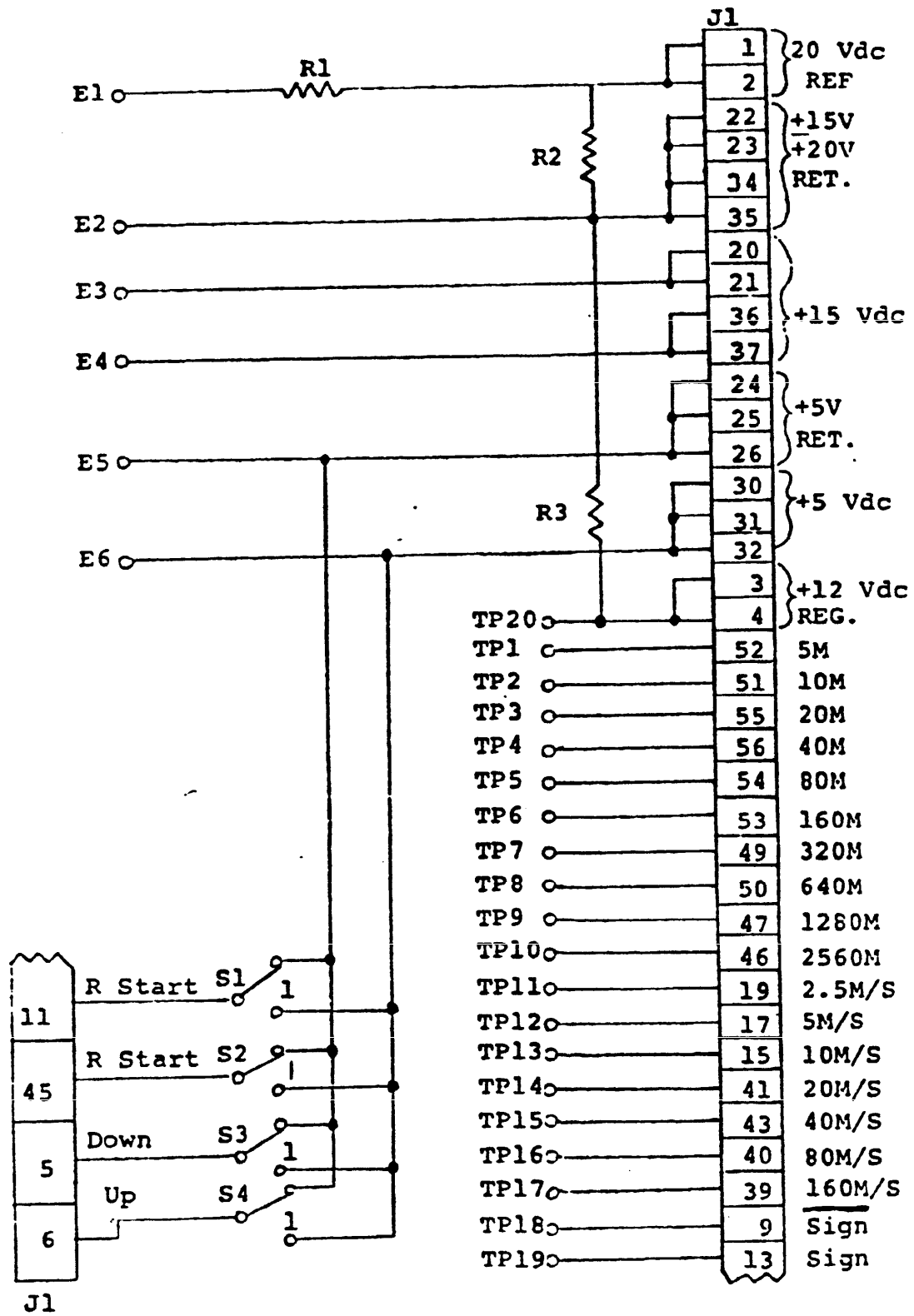


FIGURE 2. Test fixture setup.

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4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz. After the application of all the shock impulses, the range and range rate generator shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with procedure III, Method 507 of MIL-STD-810; except that in Step 7 checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical.

4.7.2.1 Continuity. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.7.2.2 Resistance. Resistance shall be measured using an ohmmeter, (item 6, Table VI), for compliance to the requirements of 3.4.2

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4.7.3 Performance.

4.7.3.1 Conditions of operation. Unless otherwise specified, all tests in 4.7.3 shall be performed with the D.C. Power Supply connections illustrated in Figure 1. The test fixture (item 5, Table VI) shall be connected to the applicable terminals of the range and range rate generator as illustrated in Figure 2. Power supplies (1, 2, and 3, Table VI) shall be connected to the designated terminals of the test fixture in compliance with the test setup of Figure 1. All power supplies shall be adjusted to meet the voltage limits given in Figure 1.

4.7.3.2 Output pin 3. with the conditions of 4.7.3.1 applied, a voltmeter (item 7, Table VI), shall be used to measure the voltage between TP20 (+) and E2(-) on the test fixture. The voltage shall be checked for compliance to the requirements of 3.5.3.

4.7.3.3 Range output. With the conditions of 4.7.3.1 applied to S1 through S4 set to "O" position, connect charnel 1 of the oscilloscope (item 4, Table VI) to TP1 of test fixture and the return to terminal ES of the test fixture. Determine that a square wave is present at these terminals and that it complies to the amplitude requirements of 3.5.4. In like reamer, apply the oscilloscope in turn to TP2 through TP1O with the return to E5 terminal. Determine that the amplitude requirements of 3.5.4 are met at each terminal, In like reamer, connect channel B of the frequency counter (item 8, Table VII) to TP1 through TP10 and verify the period requirements specified for each test point of Table I.

4.7.3.4 Range rate output. With the conditions of 4.7.3.1 applied and S1 through S4 set to "O" position, connect channel 1 of the oscilloscope (item 4, Table VI) to TP1l of the test fixture and return to terminal ES of the test fixture. Determine that a square wave is present at these terminals and that it complies to the amplitude requirements of 3.5.5. In like manner, apply the oscilloscope in turn to TP12 through TP17 with the return to ES terminal. Determine that the amplitude requirements of 3.5.5 are met at each terminal. In like manner, connect charnel B of the frequency counter (item 8, Table VI) to TP11 through TP17 and verify the period requirement specified for each test point of Table II.

4.7.3.5 Range count signals.

4.7.3.5.1 Up count range. With the conditions of 4.7.3.1 applied, set S1 through S4 on the test fixture to step 1 of Table III. Connect channel and sync of the oscilloscope (item 4, Table VI) to TP1 of the test fixture. Connect channel 2 of the oscilloscope to TP2 of the test fixture. Connect channel B of the frequency counter (item 8, Table VI) to TP2 of the test fixture and connect channel C of the counter to TP1 of the unit under test. Set the counter function to time B-C and set charnel B trigger to

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(+) and channel C trigger to (-). Toggle S1 on the test fixture to "0" and back to "1". observing the oscilloscope, the signals shall be out of phase until the counter stops counting, then in phase until the run is over. The counter shall verify-the time requirement of 3.5.6.1 for the first half of the run. Set the counter channel "C" trigger to (+) and repeat S1 toggle. The counter shall verify the time requirement of 3.5.6.1 for an entire run.

4.7.3.5.2 Down count range. With the conditions of 4.7.3.1 applied, set S1 through S4 on the test fixture to step 3 of Table III. Connect channel 1 and sync of the oscilloscope to TP1 of the test fixture. Connect channel 2 of the oscilloscope to TP2 of the test fixture. Connect charnel B of the frequency counter to TP2 of the test fixture. Connect channel C to TP1 on the unit under test. Set the counter function to time B-C and set channel B trigger to (+) and channel C trigger to (-). Toggle S1 on the test fixture to "0" then back to "1" position. observing the oscilloscope, the signals must be *out* of phase until the counter stops counting, then in phase until the run Is over. The counter shall verify the time requirement of 3.5.6.2 for the first half of the run. Set the counter charnel C trigger to (+) and repeat S1 toggle. The counter shall verify the time requirement of 3.5.6.2 for an entire run.

4.7.3.6. Range rate count signals.

4.7.3.6.1 Up count range rate. with the conditions of 4.7.3.1 applied, set S1 through S4 on the test fixture to Step 2 of Table III. Connect channel 1 and sync of the oscilloscope to TP11 of the test fixture. Connect channel 2 of the oscilloscope to TP12 of the test fixture. Connect charnel B of the frequency counter to TP12 of the test fixture and connect charnel C to TP1 of the unit under test. Set the counter function to time B-C and set channel B trigger to (+) and channel C trigger to (-). Toggle S2 on the test fixture to "0" then back to "1" position. Observing the oscilloscope, the signals must be out of phase until the counter stops counting, then in phase until the run is over. The counter shall verify the time requirement of 3.5.7.1 for the first half of the run. Set the counter charnel C trigger to (+) and repeat first half of the run. Set the counter channel C trigger to (+) and repeat S1 toggle. The counter shall verify the time requirement of 3.5.7.1 for an entire run.

4.7.3.6.2 Down count range rate. with the conditions of 4.7.3.1 applied, set S1 through S4 on the test fixture to step 4 of Table III. Connect channel 1 and sync of the oscilloscope to TP11 of the test fixture. Connect channel 2 of the oscilloscope to TP12 of the test fixture. Connect channel B of the frequency counter to TP12 of the test fixture and connect charnel C to TP1 of the unit under test. Set the counter function to time B-C and set channel B trigger to (+) and charnel C trigger to (+). Toggle S1 on the test fixture to "0" and back to "1" position. Observing the oscilloscope, the signals shall be in phase until the counter stops counting, then out of phase until the run is over. The counter

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shall verify the time requirement of 3.5.7.2 for the first half of the run. Set the counter charnel C to trigger (-) and repeat S1 toggle. The counter shall verify the time requirements of for an entire run.

4.7.3.7 Sign, sign output. With the conditions of 4.7.3.1 applied, set S1 through S4 switches on the test fixture to step 4 of Table III. Connect the frequency counter channel B to TP11 of the test fixture and connect channel C of the counter to TP1 of the unit under test. Set the counter to time B-C and set channel B trigger to (+) and channel C trigger to (+). Connect the digital voltmeter (item 7 Table VI) to TP18 of the test fixture and connect the digital voltmeter (item 7 Table VI) to TP19 of the test fixture. Toggle S2 on the test fixture to "O" and back to 'l" position. The voltmeters shall verify the requirements of 3.5.8 for in phase signals until the counter stops counting, then shall verify the requirements of 3.5.8 for out of phase signals till the end of the run. Set the S1 through S4 switches on the test fixture to step 2 of Table III. Set the counter channel C trigger to (-). Toggle S2 and the voltmeters shall verify the requirements of 3.5.8 for out of phase signals until the counter stops counting, then shall verify the requirements of 3.5.8 for in phase signals until the end of the run.

4.8 Soldering. Soldering process shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.7.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PSl1747401. The level of protection shall be as specified in the procurement document.

60 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The range and range rate generator is used to provide range and range rate signals to the SCG Servo Test panel to locate defective portions of the mechanical and electronic servo assembly.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD) Form 1423 in the contract.

6.4 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC .

6.5 Subject term (key word) listing.

SCG Servo Test Panel

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A741)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-48584B (AR)	14 June 1991

3. DOCUMENT TITLE
CIRCUIT CARD ASSEMBLY, 11747101 (RANGE AND RANGE RATE GENERATOR)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (*Include Zip Code*) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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